Mid sem 3 practice questions

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📘 **PART A – Networking (15 Marks)**

Short Answer Questions (1–2 Marks Each)

1. Define a computer network.
2. What is a network topology? Name any two topologies.
3. Differentiate between LAN and WAN.
4. What is the difference between serial and parallel transmission?

Medium Answer Questions (3–4 Marks Each)

5. Explain any **two types of network topologies** with diagrams.
6. Write a short note on **network classification** with examples.
7. Compare **OSI Model** and **TCP/IP Model** (any 3 points).
8. Explain any **three layers of the OSI model**.

Long Answer (5 Marks)

9. Draw and explain the **OSI model** with its seven layers.
10. Explain all **types of network topologies** with advantages and disadvantages.

📘 **Networking – Extended Question Set (for 15 Marks)**

□ Section A: Very Short Answer (1 Mark each)

1. Define topology in one line.
2. What is meant by parallel transmission?
3. What is TCP/IP?
4. Name any two transmission modes.
5. Expand OSI and TCP/IP.

□ Section B: Short Answer (2 Marks each)

6. State two differences between serial and parallel transmission.
7. Write any two differences between LAN and WAN.

8. Name and define any two types of network topology.
9. Write the function of the **Transport Layer** in OSI.
10. What is a Personal Area Network (PAN)? Give an example.

□ Section C: Medium Answer (3–4 Marks each)

11. Explain any **three network topologies** with diagrams.
12. Compare the OSI and TCP/IP models (any 4 points).
13. Write a short note on the **classification of networks** with a real-world example.
14. Differentiate between LAN, MAN, and WAN.
15. Explain **serial and parallel transmission** with a labeled diagram.
16. Describe **any four layers** of the OSI model and their functions.

□ Section D: Long Answer (5 Marks each – Choose Any One in Exam)

17. Draw the **OSI model** and explain all seven layers in proper sequence.
18. What is network topology? Explain **Bus, Star, and Mesh topologies** with diagrams and advantages/disadvantages.
19. Explain the **TCP/IP model** and compare it with the OSI model.
20. What are the main types of networks? Explain with **suitable examples** and **uses**.

□ BONUS: Sample Questions Teachers Often Ask

1. Which layer in the OSI model handles error detection?
2. In which topology do all computers connect to a central hub?
3. How many layers are there in the TCP/IP model? Name them.
4. Why is serial transmission preferred over parallel in long-distance communication?